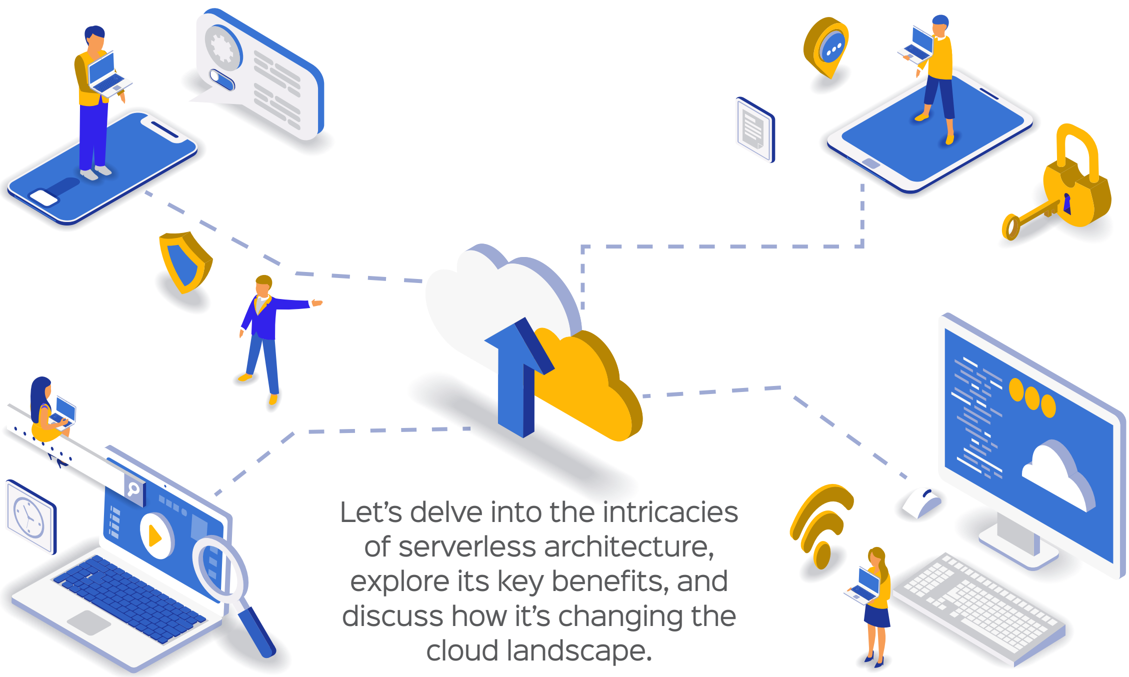
Serverless Computing:
What It Is and How It's Changing the Cloud Landscape



Let's delve into the intricacies of serverless architecture, explore its key benefits, and discuss how it's changing the cloud landscape.

Imagine you're building an automated text messaging app that sends personalised messages to users based on their preferences or behaviour. As the app gains popularity, the number of users and messages sent will increase dramatically. In a traditional server-based architecture, you would need to manually scale your servers to handle the increased load, which can be time-consuming, expensive, and prone to errors.

With serverless computing, however, the cloud provider automatically scales the necessary resources to handle the increased traffic. This means that your app can seamlessly handle peak loads without any downtime or performance degradation.

Serverless computing is not about the absence of servers; it's about liberating developers from the burden of infrastructure, letting them focus on what truly matters—building and innovating.

What is serverless computing?

Serverless computing is a cloud computing model where developers can build, run, and manage applications without having to provision or manage servers. That means developers don't need to worry about infrastructure details like hardware, operating systems, or scaling.

Table 1 highlights the key differences between traditional computing and serverless computing.

Types of serverless computing

Function as a service (FaaS): In FaaS, developers can write and deploy individual functions that execute in response to specific events. These functions are stateless, and the cloud provider manages scaling, execution, and maintenance of the infrastructure.

Examples: AWS Lambda, Azure Functions, Google Cloud Functions

Backend as a service (BaaS): BaaS provides preconfigured backend services like databases, authentication, storage, and APIs. Developers focus on frontend logic, while the cloud provider handles the backend infrastructure.